

Wi-Fi Tool And Indoor Location TRACKING DEVICE

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Modern Wi-Fi signals can do more than transfer data. These radio waves can be analysed using channel state information (CSI) to detect movement, estimate location, track activity, and even sense physiological signals. By extracting fine-grained amplitude and phase data across multiple sub-carriers, CSI can potentially detect people, including through some obstructions; estimate their distance and approximate position relative to a router; track movement; recognise activities and falls; and monitor vital

signs such as breathing rate. This turns ordinary Wi-Fi infrastructure into a contactless sensing platform for indoor monitoring, security, health, and navigation.

The design is centred on this capability and uses a minimal standalone circuit based on the ESP32-S3. Real-time CSI and signal-strength information are displayed on an OLED screen, with an option for a colour SPI LCD. Four switches and a potentiometer provide menu selection and control. The ESP32-S3 can be programmed through USB-C

solder pads on the board or pre-programmed using a dedicated programmer jig before being mounted on the PCB.

The hardware is designed to accommodate the ESP32-S3 directly as an SMD component. The PCB includes footprints for commonly available displays and modules, so the

display can be changed without redesigning the board. Internal PCB routing is preconfigured to support either an OLED or colour SPI LCD. The device also supports indoor navigation by providing real-time distance and approximate position information relative to an access point, along with continuous Wi-Fi scanning and passive monitoring and analysis.

These features make it a compact and portable Wi-Fi sensing and analysis tool for indoor tracking, monitoring, navigation, and diagnostic applications. Fig. 1 shows the author's prototype of the Wi-Fi tool. The components required to build the device are listed in the Bill of Materials table.

Circuit and working

Fig. 2 shows the complete circuit of the Wi-Fi tool and indoor location tracking device. Power is supplied through the USB-C solder-pad connector, which provides 5V to the AMS1117, which is a 3.3V, 1A LDO regulator (IC1). The ESP32-S3-WROOM (IC2) is the main

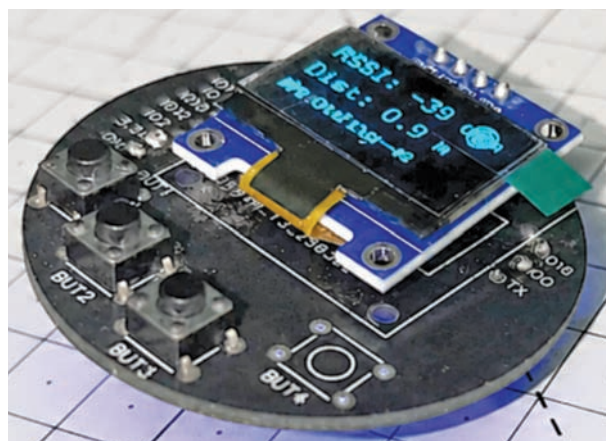


Fig. 1: Author's prototype of the Wi-Fi tool

BILL OF MATERIALS

Name	Designator	Footprint	Manufacturer part	Manufacturer	Qty
100nF	C1	C0805	CC0805KRX7R9BB104	YAGEO	1
22μF	C2,C3	C0805	CL21A226MAYNNNE	Samsung	2
10μF	C4	C0805	CL21A106KAYNNNE	Samsung	1
AMS1117-3.3V, 1A LDO regulator	IC1	SOT-223-4_L6.5-W3.5-P2.30-LS7.0-BR	AMS1117-3.3	UMW	1
ESP32-S3-WROOM	IC2	WIRELM-SMD_ESP32-S3-WROOM-1	ESP32-S3-WROOM-1-N8R8	Espressif	1
TS3325A 250GF 025	SW1	KEY-SMD_4P-L3.1-W3.1-P2.00-LS4.0	TS3325A 250GF 025	Shou Han	1
TSC005A2518A	SW2, SW3, SW4, SW5	SW-SMD_4P-L5.2-W5.2-P3.70-LS6.4	TSC005A2518A	BZCN	4
USB-C solder-pad connector	U1	USB-C solder-pad connector	—	—	1
SSD1306 OLED	U2	SSD1306-OLED-128X64-I2C	—	—	1
50kΩ potentiometer	U3	RES-ADJ-SMD_RA1010N-1H-2016-B50K-000	RA1010N-1H-2016-B50K-000	G-Switch	1
GC9A01	SPI	GC9A01 ROUND	19192	Waveshare	1